

SARAVANAVEL E

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Summary

AI & Data Science undergraduate with experience building deployable ML systems, backend APIs, and real-time AI pipelines. Strong in Python, data structures, and problem solving with focus on scalable system design and production-oriented AI applications.

Technical Skills

Languages: Python, Java, SQL

Relevant Coursework: Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning

Data Analysis: NumPy, Pandas

Machine Learning: Scikit-learn, XGBoost

Deep Learning: TensorFlow, Keras, PyTorch, CNNs, RNNs, OpenCV, Transformers (HuggingFace)

LLM/GenAI: LangChain, LangGraph, LlamaIndex, RAG Pipelines, Multi-Agent Workflows, Prompt Orchestration

Backend & APIs: Flask, FastAPI, REST APIs, Streamlit

Frontend: React.js, HTML, CSS

Databases & Data Engineering: PostgreSQL, Apache Kafka, Snowflake

MLOps & Cloud: Docker, MLflow, Git, GitHub, AWS EC2 Basics, GCP basics

Projects

- Responsible AI Testing & Bias Detection Framework for LLMs**  GitHub May 2026
Tech Stack: Python, HuggingFace Transformers, LangChain, Great Expectations, Streamlit, Docker
Developed an MVP framework for evaluating LLM responses on fairness, hallucination, and privacy-related risks. Built automated evaluation pipelines using LangChain and HuggingFace models with validation workflows for responsible AI testing. Implemented a Streamlit dashboard for monitoring evaluation metrics and generating mitigation insights.
- Topaz-Inspired Agentic AI-Powered Intelligent Ticket Resolution System**  GitHub May 2026
Tech Stack: Python, LangGraph, LangChain, FastAPI, Streamlit, PostgreSQL, ChromaDB, Docker, LangSmith
Built an MVP agentic AI-based ticket resolution system inspired by enterprise ITSM platforms using LangGraph and RAG pipelines. Developed autonomous workflows for ticket classification, diagnosis, resolution, and escalation with human-in-the-loop approval mechanisms. Integrated FastAPI backend, Streamlit dashboard, PostgreSQL persistence, and LangSmith-based observability for evaluation and monitoring.
- AI-Driven Adaptive Traffic Optimization and Emergency Corridor System**  GitHub Apr 2026
Tech Stack: Python, OpenCV, TensorFlow, CNN, YOLO, Flask, CrewAI
Designed a multi-agent AI system using YOLO-based vehicle detection and CrewAI for adaptive traffic signal control. Built a real-time pipeline for congestion detection, prediction, and signal optimization using Flask-based APIs. Automated emergency vehicle prioritization and dynamic signal switching, improving traffic decision efficiency by 20–25%.
- Data Analytics Model to Reduce Rejection Rates in Small Industries**  GitHub Feb 2026
Tech Stack: Pandas, NumPy, Matplotlib, Seaborn, Linear Regression, Streamlit
Developed a data analytics system using Pareto Analysis and Statistical Process Control (SPC) to identify high-impact defect causes in manufacturing pipelines. Built analytical workflow to process production data and detect bottlenecks, enabling reduction in rejection patterns by identifying high-impact defect contributors. Implemented a Streamlit dashboard for real-time monitoring and decision support.

Experience

InternPro – Artificial Intelligence Intern Jul 2025 – Sep 2025
Built preprocessing and feature engineering pipelines for weather forecasting models using Python and Scikit-learn.

Techno Hacks Solutions Pvt. Ltd. – Data Science Intern Sep 2024 – Oct 2024
Applied data analysis and ML techniques on real-world datasets.

Education

B.Tech - Artificial Intelligence and Data Science 2023 – 2027
Karpagam College of Engineering, Coimbatore. CGPA: 8.13 / 10

Higher Secondary Education - PCBM 2021 – 2023
Everest Mariappa Nadar Higher Secondary School, Kovilpatti. Percentage: 92%